

Ahmed Hisham Gaballah

Generative AI Engineer | Agentic AI & Multi-Agent Systems | LLM Developer

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PROFESSIONAL SUMMARY

Entry-level Generative AI Engineer (graduating Aug 2026) with hands-on experience building production-ready LLM applications, multi-agent architectures, and RAG pipelines. Proficient in LangChain, LangGraph, OpenAI API, and Python. Designed supervisor agents, tool-calling sub-agents, and state graph orchestration for real-world AI workflow automation. Delivered multi-turn chatbots, NLP-powered sentiment analysis, and natural-language-to-SQL database systems. Familiar with Docker, FastAPI, Git, LangSmith, and CI/CD principles. Open to fully remote roles in the USA, UAE, and KSA.

TECHNICAL SKILLS

Generative AI & Agentic Systems: Multi-Agent Architecture, Agentic AI Workflows, Supervisor & Sub-Agent Design, Tool Calling, Human-in-the-Loop, State Graphs, Long-Term Memory, Autonomous AI Agents, Structured Outputs, Agent Memory Management

LLM Frameworks & Orchestration: LangChain, LangGraph, OpenAI API, Hugging Face Transformers, LlamaIndex (familiar), Prompt Engineering (Zero-Shot, Few-Shot, Chain-of-Thought), Model Evaluation (ROUGE, BLEU), Guardrails / Responsible AI

RAG & Vector Databases: Retrieval-Augmented Generation (RAG), ChromaDB, FAISS, Weaviate (familiar), Milvus (familiar), Pinecone (familiar), Semantic Search, Embeddings, Document Chunking, Elasticsearch (familiar)

NLP & Machine Learning: Natural Language Processing (NLP), Text Classification, Sentiment Analysis, PyTorch (familiar), Transformers Architecture, Deep Learning Fundamentals, Scikit-learn, Pandas, NumPy

Programming & Development: Python (primary), FastAPI (familiar), REST APIs, OOP, Streamlit, Async Python (familiar), C, Embedded C, Linux, Makefiles, CMake, Git, GitHub

Databases: SQLite (SQL), ChromaDB (Vector Database), Natural Language to SQL, Data Pipelines

MLOps & DevOps (Familiar): Docker, GitHub Actions, CI/CD Pipelines, LangSmith (Observability & Tracing), MLflow (familiar), Workflow Automation (n8n / Make — familiar), AI Pipeline Deployment, Prompt Versioning, MCP (Model Context Protocol — familiar)

Embedded Systems: AVR ATmega32, GPIO, UART, I2C, SPI, ADC, PWM, EEPROM, FreeRTOS (intro), Proteus

Languages: Arabic (Native), English (Professional — B2/C1)

PROJECTS

Music Store Customer Support — Multi-Agent System | *Generative AI Diploma Final Project* 2025

- Built a production-grade customer support multi-agent system for a digital music store using LangChain, LangGraph, Python, and Jupyter Notebook.
- Designed a Supervisor Agent to orchestrate and route queries to specialized sub-agents: a Music Catalog Agent (album/track/genre search & recommendations) and an Invoice Information Agent (purchase history & employee lookup).
- Implemented customer verification with human-in-the-loop interruption logic: workflow pauses and requests missing credentials (customer ID, email, or phone) before resuming.
- Integrated long-term memory store for personalized recommendations based on user preferences, and natural language querying of the Chinook SQLite database.
- Containerized the application with Docker for consistent deployment.
- Tech Stack: Python, LangChain, LangGraph, SQLite (Chinook), Google API, State Graphs, Tool Calling, Human-in-the-Loop, Docker

Conversational RAG Q&A System with PDF Upload & Chat History | *Generative AI Diploma Project* 2025

- Built a context-aware multi-turn Q&A chatbot in Streamlit capable of answering questions over multiple uploaded PDFs simultaneously.
- Implemented a full RAG pipeline using LangChain, ChromaDB vector storage, and HuggingFace embeddings for accurate knowledge retrieval.
- Integrated session memory for history-aware multi-turn dialogue; modular OOP architecture supports swappable vector store adapters.
- Tech Stack: Python, LangChain, LangGraph, ChromaDB, HuggingFace, OpenAI API, Streamlit

Student Database Management System with AI Chatbot | *Generative AI Diploma Project* 2025

- Designed a secure multi-role student information platform with admin and user dashboards and role-based access control.

- Implemented SHA-256 password hashing for authentication and integrated a Streamlit AI chatbot to query database records via natural language.
- Tech Stack: Python, SQLite, Streamlit, OOP, SHA-256 Authentication

Twitter Sentiment Analysis Application | *Personal Project*

2024

- Trained a lightweight ML classifier to perform sentiment analysis (positive, negative, neutral) on Twitter/X data with real-time ingestion.
- Preprocessed a Kaggle Twitter dataset and built a text classification pipeline achieving reliable multi-class prediction.
- Tech Stack: Python, Scikit-learn, Pandas, Kaggle Dataset

Embedded Systems Projects — ATmega32 | *Personal / Academic*

2024

- Developed low-level hardware drivers for GPIO, ADC, UART, I2C, SPI, PWM, and EEPROM on the AVR ATmega32 microcontroller.
- Built functional prototypes: smart locking system (UART), smart home automation, dual-mode stopwatch, and car parking assistance system.
- Tech Stack: Embedded C, AVR ATmega32, Proteus

PROFESSIONAL EXPERIENCE

Embedded Systems Head | *IEEE — Fayoum University Student Branch*

Oct 2024 – May 2025

- Led the embedded systems team; designed and delivered training sessions on AVR microcontrollers and C programming for 20+ students.
- Supervised and debugged student hardware projects; coordinated workshops to foster hands-on embedded development skills.
- Provided technical mentorship to junior members, improving team project completion rate and collaboration within the IEEE branch.

Engineering Intern | *El-Korimat Power Plant*

Sep 2022 – Oct 2022

- Gained practical exposure to combined-cycle power generation systems and pump station operations.
- Assisted senior engineers with diagnostics, maintenance procedures, and technical documentation of mechanical systems.

EDUCATION

B.Eng. in Mechatronics Engineering | *Faculty of Engineering, Fayoum University*

Aug 2021 – Jun 2026 (Expected)

GPA: 3.71 / 4.0 | Relevant Coursework: Mechanics, Electronics, Programming, Automation, Advanced Mathematics

CERTIFICATIONS & DIPLOMAS

Generative AI Diploma | *Edges for Training Academy*

Jun 2025 – Aug 2025

- AI foundations: Machine Learning, Deep Learning, Transformers, LLMs, Prompt Engineering (zero-shot, few-shot, CoT), RAG.
- Hands-on projects: Multi-Agent Customer Support System (Final Project), PDF-based RAG chatbot, database-driven chatbot, LangChain & LangGraph pipelines.
- Tools: Hugging Face, OpenAI API, ChromaDB, Streamlit, LangGraph, State Graphs.

Embedded Systems Diploma | *Edges for Training Academy*

Jun 2024 – Nov 2024

- Embedded C, AVR microcontrollers, driver development (GPIO, UART, I2C, ADC, PWM, EEPROM), FreeRTOS (intro), hardware interfacing.
- Hands-on labs with ATmega32 including driver implementation and real-world embedded system applications.